

Madhan Karthikeyan

+91 93619 51624 | madhankrthik@gmail.com | linkedin.com/in/madhan-karthikeyan | github.com/madhan-karthikeyan

SUMMARY

Computer Science undergraduate (VIT + IIT Madras) with hands-on experience across ML systems and full-stack backend engineering – computer vision, embedding-based retrieval, time-series forecasting, REST API design, and async task pipelines; skilled at building end-to-end systems from data curation through optimized production deployment.

EDUCATION

Vellore Institute of Technology (VIT) <i>B.Tech in Computer Science & Engineering – CGPA: 8.80/10</i>	Vellore, Tamil Nadu Aug 2023 – Present
Indian Institute of Technology, Madras <i>B.S. in Data Science and Applications</i>	Online Sep 2023 – Present

WORK EXPERIENCE

AI Engineer Intern <i>National Informatics Centre Govt. of India</i> - Architected a triage-to-resolution pipeline integrating a vision-language model for structured issue extraction with a vector similarity index over 150+ documents (4GB), cutting resolution time from >24h to under 1 minute. - Engineered the document ingestion service with chunking, embedding generation, and metadata-filtered cosine retrieval, enabling sub-second search over 4GB without dedicated search infrastructure.	May 2025 – Jul 2025 Chennai, Tamil Nadu
Computer Vision Intern <i>GloballyGI</i> - Engineered a multi-object food detection pipeline (YOLOv8n + D-FINE), achieving 80–85% mAP across 50 food classes on a custom-curated 5,000-image annotated dataset. - Designed class-balancing and targeted augmentation pipelines for 50 categories, reducing per-class mAP variance by ~35% and improving generalization on occluded, multi-object scenes. - Reduced model inference latency by ~25% through lightweight architecture selection (YOLOv8n) and post-training optimization, enabling real-time detection suitable for production deployment.	Jan 2025 – Apr 2025 Remote

PROJECTS

MediCore – Hospital Management Platform <i>Full-Stack Distributed System</i> - Architected a multi-role hospital platform on Flask + Vue.js with JWT-secured RBAC across ~40 REST APIs, supporting independent admin, doctor, and patient workflows across a 3-tier user hierarchy. - Designed async task pipelines using Celery + Redis for reminder dispatch, report generation, and bulk data export – decoupling long-running jobs from the HTTP request cycle and eliminating request timeout failures. - Implemented event-driven Redis cache invalidation maintaining <150ms p95 API latency under concurrent multi-role load; integrated Razorpay with idempotent transaction handling to eliminate duplicate payment risk.	Flask, Vue.js, Celery, Redis, PostgreSQL, Razorpay, JWT
Satellite Image Search System <i>Embedding-Based Geospatial Retrieval</i> - Built a semantic retrieval system over 10,000+ geo-tagged satellite images using RemoteCLIP embeddings and ChromaDB vector search, achieving <500ms end-to-end query latency for geospatial discovery across multispectral datasets. - Designed preprocessing workflows for 4-band multispectral GeoTIFF data; applied DBSCAN with Haversine metric to cluster retrieval results into interpretable geographic regions. - Exposed retrieval and clustering via a FastAPI backend with bounding-box query support and 3D map visualization using Cesium ion.	RemoteCLIP, ChromaDB, DBSCAN, FastAPI, Cesium
Responsible Financial AI – RAG System <i>Full-Stack Retrieval-Augmented Generation</i> - Built a production-grade RAG pipeline using LlamaIndex and ChromaDB for context-grounded financial Q&A, with structured document ingestion, embedding indexing, and cosine similarity retrieval to reduce hallucinations. - Delivered a FastAPI backend and ReactJS frontend serving explainable, source-cited responses with <2s end-to-end latency across a 1,000+ document knowledge base.	LlamaIndex, ChromaDB, FastAPI, ReactJS
DriftBench-TS – Time-Series Benchmarking Framework <i>ML Systems + Software Engineering</i> - Built a configuration-driven benchmarking framework supporting 5+ forecasting algorithms and 8+ concept drift detectors with plug-and-play integration, rolling-window simulation, and automated retraining strategies. - Exposed experiment orchestration via a Flask REST API and React dashboard providing real-time metric visualization across 6 benchmark datasets, reducing per-experiment setup time from hours to minutes.	Python, Flask, React, LightGBM, Configuration-Driven Pipelines
Cluster-Aware Semantic Retrieval & Caching System <i>ML Retrieval System</i> - Partitioned embedding space with Gaussian Mixture Models to scope FAISS search within cluster regions, reducing average query latency by ~30% over naïve global search. - Implemented a cluster-indexed semantic cache enabling result reuse for similar queries; the full pipeline (embedding → GMM routing → cache → FAISS fallback) sustained a ~35% cache hit rate under repeated-query workloads, served via FastAPI.	GMM, FAISS, FastAPI, Python

SKILLS

ML & Modeling: PyTorch, Scikit-learn, LightGBM, YOLOv8, Time-Series Forecasting, Feature Engineering

CV & Retrieval: CLIP / RemoteCLIP, FAISS, ChromaDB, Vector Search, Embedding Pipelines, Object Detection

ML Systems: GMM Clustering, DBSCAN, RAG Pipelines, Semantic Caching, LlamaIndex

Backend: Python, Flask, FastAPI, REST API Design, JWT Authentication, Celery, Redis, Distributed Task Queues

Frontend: ReactJS, Vue.js, TanStack Router, Tailwind CSS

Databases: PostgreSQL, MongoDB, ChromaDB, Redis

Systems & DevOps: Docker, AWS (EC2, S3), Git, Event-Driven Architecture, Async Pipelines

Programming & Data: Python (Pandas, NumPy, Matplotlib, Seaborn), SQL, JavaScript

CERTIFICATIONS

Microsoft Certified: Azure AI Engineer Associate

IIT Madras: Advanced Certificate in Machine Learning & Data Science